26 Here are the first four terms of an arithmetic sequence.

5 11 17 23

Write down an expression, in terms of n, for the nth term of the sequence.

First differences: +6, +6, +6 → start with 6n.

1 mark
Use the constant difference

6n gives 6, 12, 18, 24: each is 1 too high.

nth term = 6n − 1

1 mark
Adjust by −1

Check: n = 1 gives 5; n = 4 gives 23.

6n − 1

(Total for Question 26 is 2 marks)